Semantic Error Categorization and Interactive Visualization using RAG-based Query Intelligence

Abstract

This project implements a Retrieval-Augmented Generation (RAG) based intelligent system for semantic error categorization and interactive visualization. The system classifies incoming issue queries into predefined business categories using embedding-based semantic similarity. It also retrieves similar historical issues and displays category-wise query distribution through an interactive bubble chart dashboard.

1. Problem Statement

Organizations receive large volumes of textual error reports and service tickets. Manual categorization is inefficient and error-prone. This project aims to automate error classification using semantic similarity and provide an interactive dashboard for real-time monitoring.

2. System Architecture

- Embedder: Converts text queries into 384-dimensional vector embeddings.
- Semantic Categorizer: Matches query embeddings with predefined category embeddings.
- Vector Store: Stores historical issue embeddings and metadata.
- RAG Pipeline: Combines categorization and similarity retrieval.
- FastAPI Backend: Exposes REST APIs for ingestion and querying.
- Streamlit Dashboard: Visualizes category counts using bubble chart.

3. Core Logic Flow

- User submits a query.
- Query is converted into embedding vector.
- Cosine similarity is computed with stored category embeddings.
- Best matching category is selected.

- Top-K similar past issues are retrieved from vector store.
- Category count is incremented for visualization.
- Results are returned via API and reflected in dashboard.

4. Technologies Used

- Python 3.10+
- NumPy for vector operations
- Sentence Transformers for embeddings
- FastAPI for backend API
- Streamlit for interactive dashboard
- Uvicorn ASGI server

5. Results

The system successfully classifies queries into Finance, IT, Reporting, Compliance, and Customer Support categories. The dashboard dynamically updates bubble sizes based on category frequency, enabling real-time monitoring of issue distribution.

6. Conclusion

This project demonstrates a scalable semantic intelligence system using RAG architecture. It combines embedding-based classification with retrieval mechanisms and interactive visualization. The modular design allows further enhancements such as LLM integration, database persistence, and real-time enterprise deployment.